Geo-Spatial Clustering

February 23, 2019

1 Geo-Spatial Clustering

```
In [1]: #Import required libraries

        import pandas as pd
        import numpy as np
        import re
        %matplotlib inline
        import numpy as np
        import pysal as ps
        import random as rdm
        from pysal.viz.splot import mapping as maps
        from pylab import *
        import geopandas as gpd
```

```
C:\ProgramData\Anaconda3\lib\site-packages\pysal\model\spvcm\abstracts.py:10: UserWarning: The
  from .sqlite import head_to_sql, start_sql
```

1.0.1 Read csv file consisting of demographic data for India

Data used below is downloaded from https://www.kaggle.com/danofer/india-census. This data is said to be demograpic data from 2011 census

```
In [2]: data = pd.read_csv("india-districts-census-2011.csv")
        data.head()
```

```
Out[2]:    District code         State name District name  Population    Male  Female  \
        0              1  JAMMU AND KASHMIR       Kupwara      870354  474190  396164
        1              2  JAMMU AND KASHMIR        Badgam      753745  398041  355704
        2              3  JAMMU AND KASHMIR   Leh(Ladakh)      133487   78971   54516
        3              4  JAMMU AND KASHMIR        Kargil      140802   77785   63017
        4              5  JAMMU AND KASHMIR         Punch      476835  251899  224936

           Literate  Male_Literate  Female_Literate    SC         ...          \
        0    439654         282823           156831  1048         ...
        1    335649         207741           127908   368         ...
```

```
2     93770          62834            30936   488         ...
3     86236          56301            29935    18         ...
4    261724         163333            98391   556         ...

   Power_Parity_Rs_90000_150000  Power_Parity_Rs_45000_150000  \
0                            94                           588
1                           126                           562
2                            46                           122
3                            27                           114
4                            78                           346

   Power_Parity_Rs_150000_240000  Power_Parity_Rs_240000_330000  \
0                             71                            101
1                             72                             89
2                             15                             22
3                             12                             18
4                             35                             50

   Power_Parity_Rs_150000_330000  Power_Parity_Rs_330000_425000  \
0                            172                             74
1                            161                             96
2                             37                             20
3                             30                             19
4                             85                             59

   Power_Parity_Rs_425000_545000  Power_Parity_Rs_330000_545000  \
0                             10                             84
1                             28                            124
2                             14                             34
3                              3                             22
4                              8                             67

   Power_Parity_Above_Rs_545000  Total_Power_Parity
0                            15                1119
1                            18                1066
2                            17                 242
3                             7                 214
4                            12                 629

[5 rows x 118 columns]
```

1.0.2 Group the Data Statewise

We have grouped the data statewise as the shape file of India which is used is at the state-level. Although we can used the shapefile wih district level informaion if we want the analysis to be done at district level.

```
In [3]: data1 = data.groupby(by='State name').sum()
```

In [4]: data1.head()

Out[4]:

```
                              District code  Population      Male    Female  \
State name
ANDAMAN AND NICOBAR ISLANDS           1917      380581    202871    177710
ANDHRA PRADESH                       12489    84580777  42442146  42138631
ARUNACHAL PRADESH                     4040     1383727    713912    669815
ASSAM                                 8451    31205576  15939443  15266133
BIHAR                                 8417   104099452  54278157  49821295

                              Literate  Male_Literate  Female_Literate  \
State name
ANDAMAN AND NICOBAR ISLANDS     294281         164377           129904
ANDHRA PRADESH                50556760       28251243         22305517
ARUNACHAL PRADESH               766005         439868           326137
ASSAM                         19177977       10568639          8609338
BIHAR                         52504553       31608023         20896530

                                    SC  Male_SC  Female_SC         ...          \
State name                                                         ...
ANDAMAN AND NICOBAR ISLANDS          0        0          0         ...
ANDHRA PRADESH                13878078  6913047    6965031         ...
ARUNACHAL PRADESH                    0        0          0         ...
ASSAM                          2231321  1145314    1086007         ...
BIHAR                         16567325  8606253    7961072         ...

                              Power_Parity_Rs_90000_150000  \
State name
ANDAMAN AND NICOBAR ISLANDS                           2741
ANDHRA PRADESH                                       34248
ARUNACHAL PRADESH                                     1486
ASSAM                                                 2095
BIHAR                                                18132

                              Power_Parity_Rs_45000_150000  \
State name
ANDAMAN AND NICOBAR ISLANDS                           6810
ANDHRA PRADESH                                       80360
ARUNACHAL PRADESH                                     4158
ASSAM                                                 6406
BIHAR                                                55371

                              Power_Parity_Rs_150000_240000  \
State name
ANDAMAN AND NICOBAR ISLANDS                            1073
ANDHRA PRADESH                                         5183
ARUNACHAL PRADESH                                       556
ASSAM                                                   834
```

```
BIHAR                                                  6168

                            Power_Parity_Rs_240000_330000  \
State name
ANDAMAN AND NICOBAR ISLANDS                              636
ANDHRA PRADESH                                          3090
ARUNACHAL PRADESH                                        396
ASSAM                                                    590
BIHAR                                                    564

                            Power_Parity_Rs_150000_330000  \
State name
ANDAMAN AND NICOBAR ISLANDS                             1709
ANDHRA PRADESH                                          8273
ARUNACHAL PRADESH                                        952
ASSAM                                                   1424
BIHAR                                                   6732

                            Power_Parity_Rs_330000_425000  \
State name
ANDAMAN AND NICOBAR ISLANDS                              405
ANDHRA PRADESH                                          3771
ARUNACHAL PRADESH                                        141
ASSAM                                                    195
BIHAR                                                    788

                            Power_Parity_Rs_425000_545000  \
State name
ANDAMAN AND NICOBAR ISLANDS                              737
ANDHRA PRADESH                                          5385
ARUNACHAL PRADESH                                        153
ASSAM                                                    207
BIHAR                                                   1273

                            Power_Parity_Rs_330000_545000  \
State name
ANDAMAN AND NICOBAR ISLANDS                             1142
ANDHRA PRADESH                                          9156
ARUNACHAL PRADESH                                        294
ASSAM                                                    402
BIHAR                                                   2061

                            Power_Parity_Above_Rs_545000  Total_Power_Parity
State name
ANDAMAN AND NICOBAR ISLANDS                              544               12084
ANDHRA PRADESH                                          6449              124166
ARUNACHAL PRADESH                                        172                6456
ASSAM                                                    202                9944
```

```
BIHAR                                                    1576                89490

[5 rows x 116 columns]
```

In [7]:
```
data1.columns
```

Out[7]:
```
Index(['District code', 'Population', 'Male', 'Female', 'Literate',
       'Male_Literate', 'Female_Literate', 'SC', 'Male_SC', 'Female_SC',
       ...
       'Hindus_Ratio', 'Muslims_Ratio', 'Sikhs_Ratio', 'Buddhists_Ratio',
       'Jains_Ratio', 'Ownership_Owned_Households_Ratio',
       'Ownership_Rented_Households_Ratio',
       'Housholds_with_Electric_Lighting_Ratio', 'Rural_Households_Ratio',
       'Urban_Households_Ratio'],
      dtype='object', length=142)
```

In [8]:
```
data1['MaleRatio'] = round(data1['Male']/data1['Population'],2)
data1['FemaleRatio'] = round(data1['Female']/data1['Population'],2)
data1['LiteracyRatio'] = round(data1['Literate']/data1['Population'],2)
data1['MaleLiteracyRatio'] = round(data1['Male_Literate']/data1['Population'],2)
data1['FemaleLiteracyRatio'] = round(data1['Female_Literate']/data1['Population'],2)
data1['Belowprimaryratio'] = round(data1['Below_Primary_Education']/data1['Population']
data1['PrimaryRatio'] = round(data1['Primary_Education']/data1['Population'],2)
data1['MiddleRatio'] = round(data1['Middle_Education']/data1['Population'],2)
data1['HigherEduRatio'] = round(data1['Higher_Education']/data1['Population'])
data1['GraduateRatio'] = round(data1['Graduate_Education']/data1['Population'])
data1['OtherEduRatio'] = round(data1['Other_Education']/data1['Population'])

data1['Age_Group_0_29_Ratio'] = data1['Age_Group_0_29']/data1['Population']
data1['Age_Group_30_49_Ratio'] = data1['Age_Group_30_49']/data1['Population']
data1['Age_Group_50_Ratio'] = data1['Age_Group_50']/data1['Population']

data1['SC_Ratio'] = data1['SC']/data1['Population']
data1['ST_Ratio'] = data1['ST']/data1['Population']
data1['Hindus_Ratio'] = data1['Hindus']/data1['Population']
data1['Muslims_Ratio'] = data1['Muslims']/data1['Population']
data1['Sikhs_Ratio'] = data1['Sikhs']/data1['Population']
data1['Buddhists_Ratio'] = data1['Buddhists']/data1['Population']
data1['Jains_Ratio'] = data1['Jains']/data1['Population']


data1['Ownership_Owned_Households_Ratio'] = data1['Ownership_Owned_Households']/data1[
data1['Ownership_Rented_Households_Ratio'] = data1['Ownership_Rented_Households']/data
data1['Housholds_with_Electric_Lighting_Ratio'] = data1['Housholds_with_Electric_Light
data1['Rural_Households_Ratio'] = data1['Rural_Households']/data1['Households']
data1['Urban_Households_Ratio'] = data1['Urban_Households']/data1['Households']
```

In [9]:
```
del data1['Male']
del data1['Female']
```

```
del data1['Literate']
del data1['Male_Literate']
del data1['Female_Literate']
del data1['Below_Primary_Education']
del data1['Primary_Education']
del data1['Middle_Education']
del data1['Higher_Education']
del data1['Graduate_Education']
del data1['Other_Education']

del data1['Age_Group_0_29']
del data1['Age_Group_30_49']
del data1['Age_Group_50']

del data1['SC']
del data1['ST']

del data1['Hindus']
del data1['Muslims']
del data1['Sikhs']
del data1['Buddhists']
del data1['Jains']

del data1['Ownership_Owned_Households']
del data1['Ownership_Rented_Households']
del data1['Housholds_with_Electric_Lighting']
del data1['Rural_Households']
del data1['Urban_Households']
```

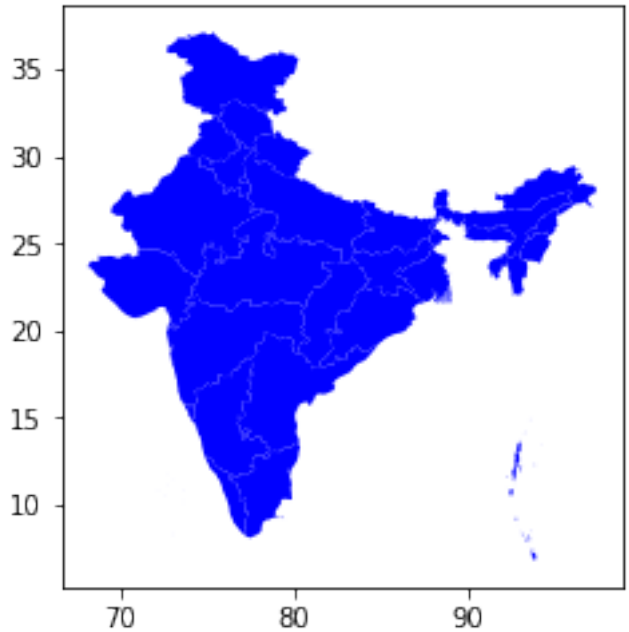
In [10]:
```
data1.columns
```

Out[10]:
```
Index(['District code', 'Population', 'Male_SC', 'Female_SC', 'Male_ST',
       'Female_ST', 'Workers', 'Male_Workers', 'Female_Workers',
       'Main_Workers',
       ...
       'Hindus_Ratio', 'Muslims_Ratio', 'Sikhs_Ratio', 'Buddhists_Ratio',
       'Jains_Ratio', 'Ownership_Owned_Households_Ratio',
       'Ownership_Rented_Households_Ratio',
       'Housholds_with_Electric_Lighting_Ratio', 'Rural_Households_Ratio',
       'Urban_Households_Ratio'],
      dtype='object', length=116)
```

1.0.3 Load Shape file of India

In [11]:
```
shp_link = './INDIA.shp'
tx = gpd.read_file(shp_link)
tx.plot(color='blue')
```

Out[11]:
```
<matplotlib.axes._subplots.AxesSubplot at 0x23a4f74b160>
```

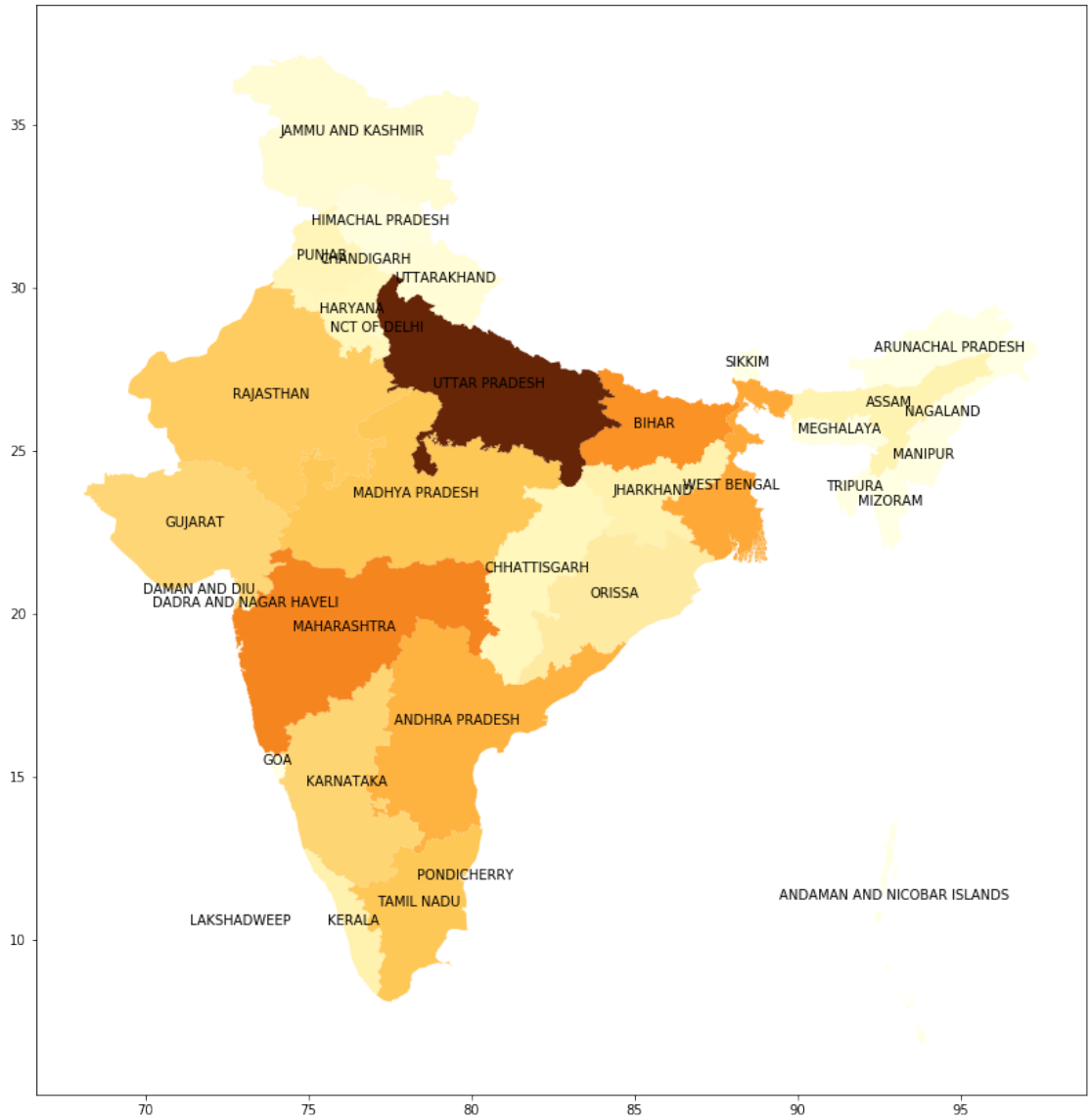



```
In [12]: tx.ST_NAME = tx.ST_NAME.str.upper()

In [13]: india = pd.merge(tx, data1, left_on='ST_NAME',right_index=True)

In [14]: india.shape

Out[14]: (35, 118)
```

1.1 Population Statewise

The plot shown below is the distribution of population in different states. Darker shades represent the higher population and ligher shapes represent lesser population comparitively.

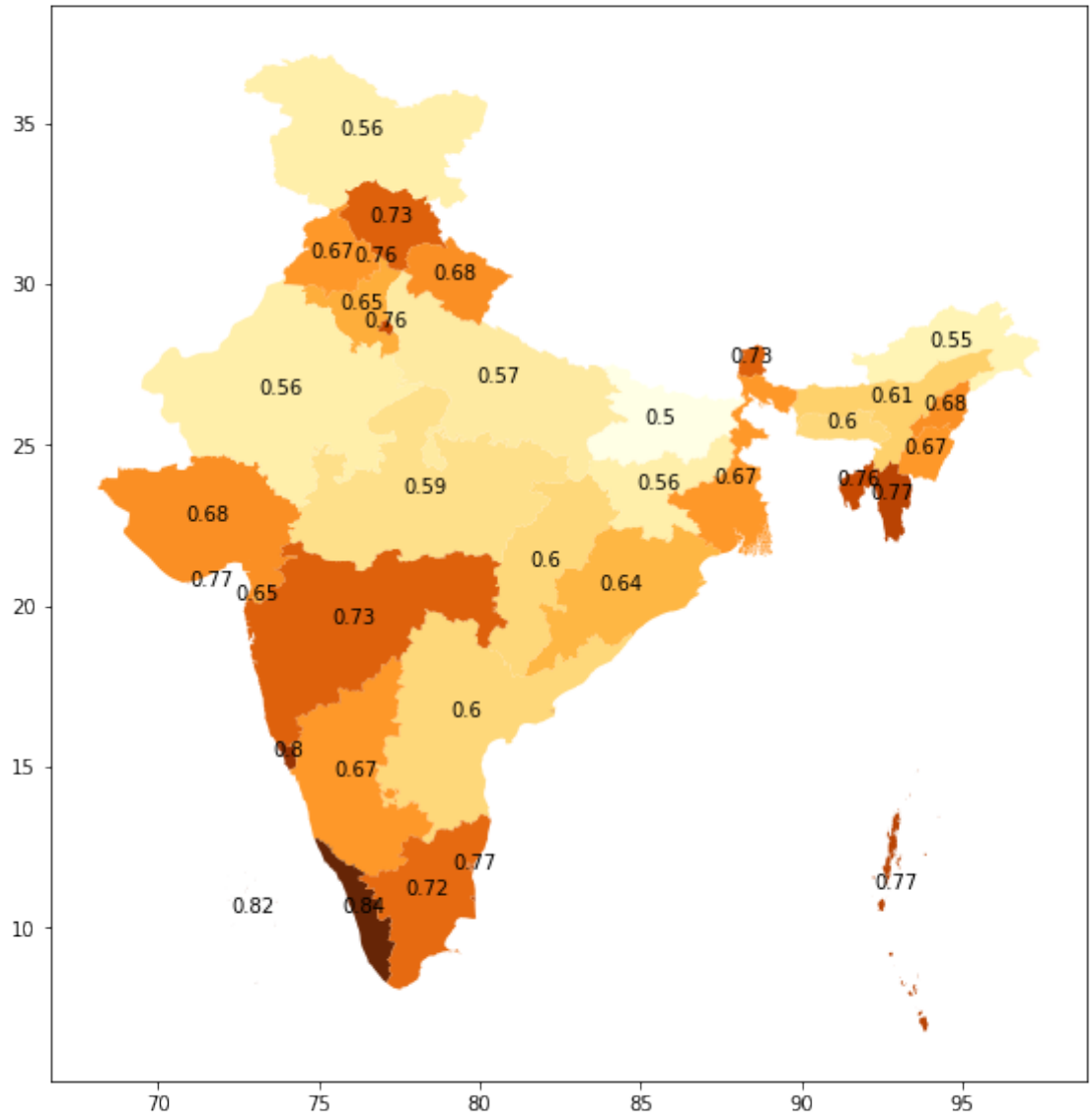
```
In [15]: f, ax = plt.subplots(1, figsize=(30, 15))
         india.plot(column='Population', cmap='YlOrBr',ax=ax)
         india.apply(lambda x: ax.annotate(s=x.ST_NAME, xy=x.geometry.centroid.coords[0], ha='
         plt.show()
```



1.1.1 Literacy Ratio Statewise

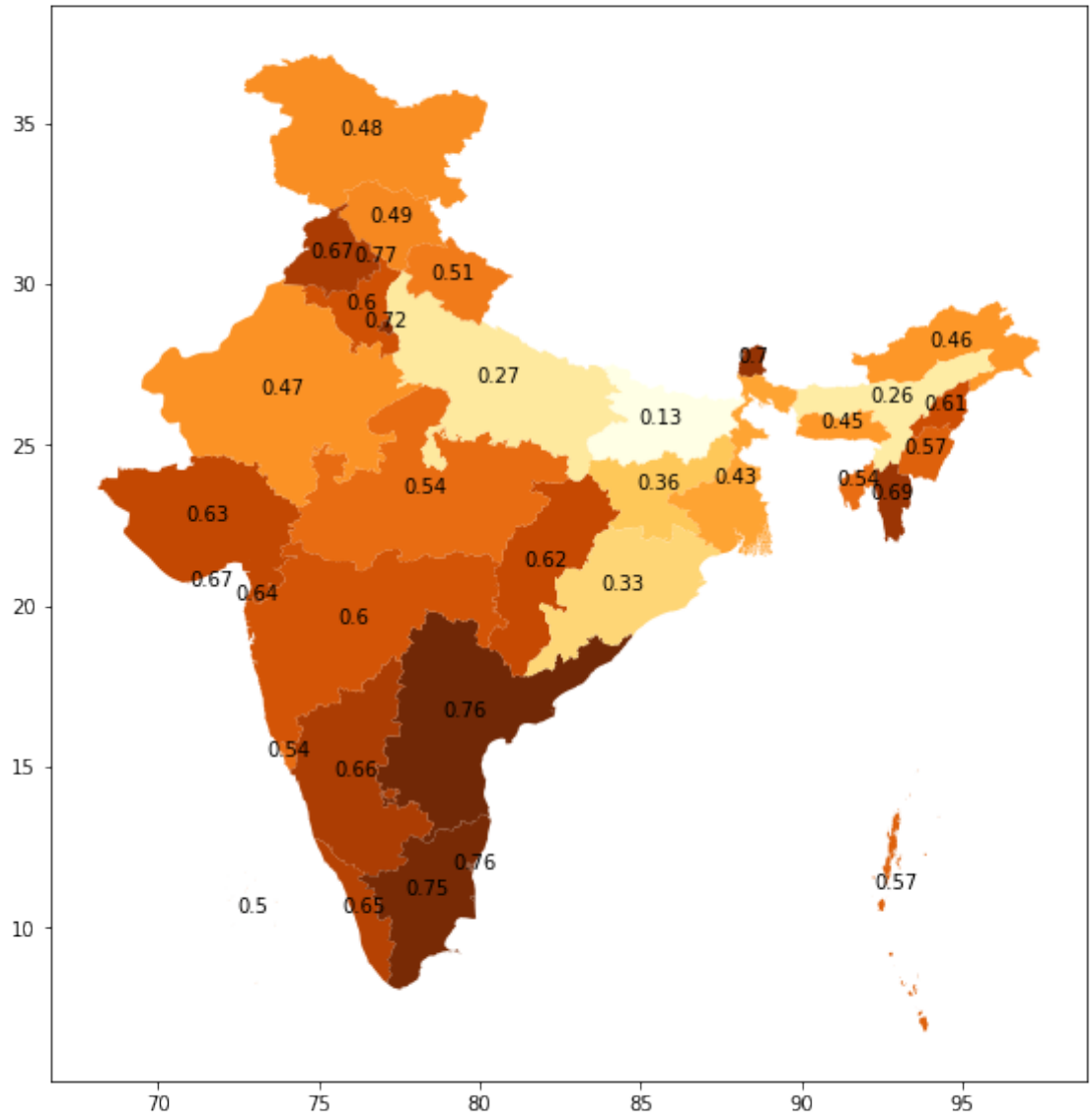
Literacy Ratio is calculated as ratio of total literate to total population in each state..

```
In [17]: f, ax = plt.subplots(1, figsize=(20, 10))
         india.plot(column='LiteracyRatio', cmap='YlOrBr',ax=ax)
         india.apply(lambda x: ax.annotate(s=x.LiteracyRatio, xy=x.geometry.centroid.coords[0]
         plt.show()
         #plt.savefig('foo.jpg', dpi=500)
```



1.1.2 Ratio of Households with Electric Supply to Total Households

```
In [32]: f, ax = plt.subplots(1, figsize=(20, 10))
         india.plot(column='Housholds_with_Electric_Lighting_Ratio', cmap='YlOrBr',ax=ax)
         india.apply(lambda x: ax.annotate(s=round(x.Housholds_with_Electric_Lighting_Ratio,2)
                                           ha='center'),axis=1)
         plt.show()
```



1.2 Spatial Similarity

In spatial autocorrelation analysis, the spatial weights are used to formalize the notion of spatial similarity. Here we will use queen contiguity:

1.2.1 Queen Weights

A commonly-used type of weight is a queen contigutiy weight, which reflects adjacency relationships as a binary indicator variable denoting whether or not a polygon shares an edge or a vertex with another polygon

```
In [19]: W = ps.lib.weights.Queen(india['geometry'])
         W.transform = 'r'
```

```
('WARNING: ', 0, ' is an island (no neighbors)')
('WARNING: ', 17, ' is an island (no neighbors)')


C:\ProgramData\Anaconda3\lib\site-packages\pysal\lib\weights\weights.py:168: UserWarning: There
  Island ids: 0, 17
  " Island ids: %s" % ', '.join(str(island) for island in self.islands))
```

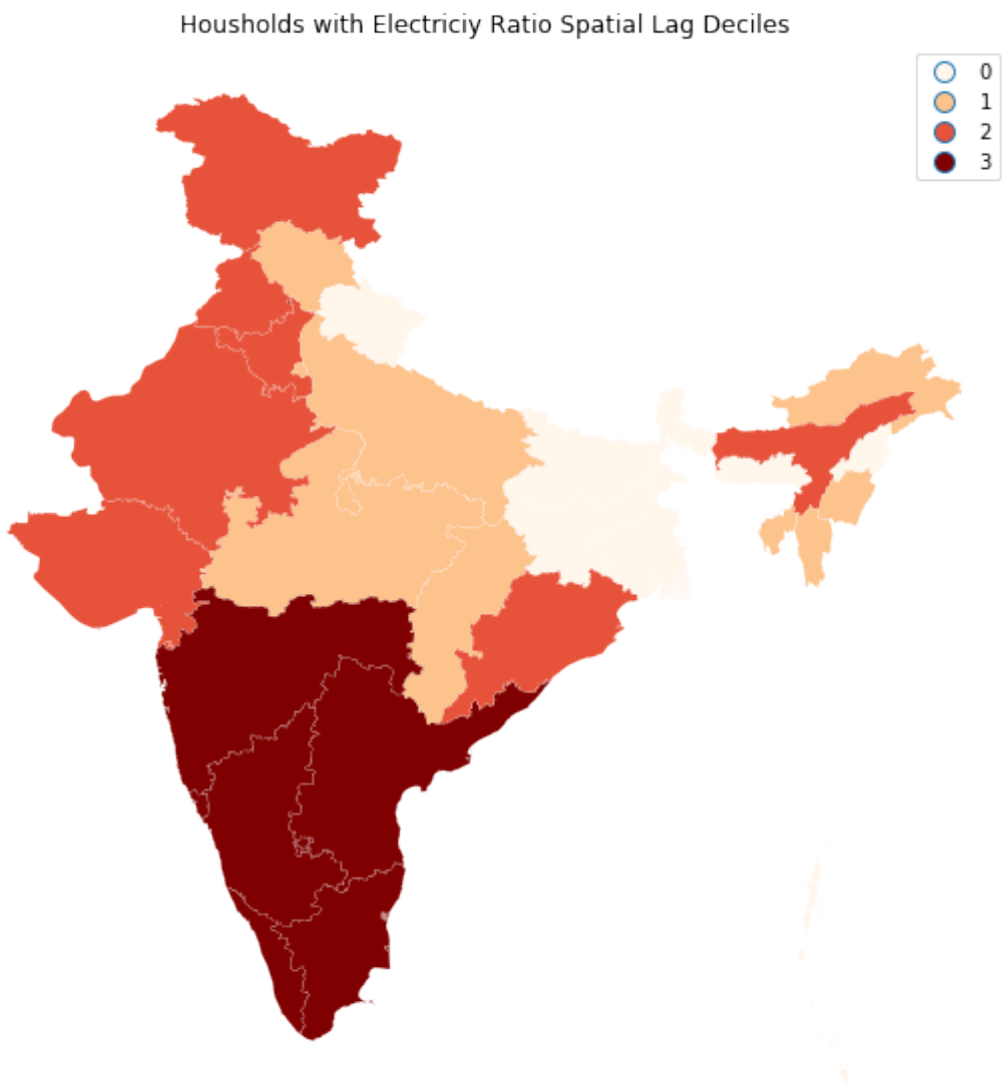
1.3 Attribute Similarity

So the spatial weight between states *i* and *j* indicates if the two states are neighbors (i.e., geographically similar). What we also need is a measure of attribute similarity to pair up with this concept of spatial similarity. The spatial lag is a derived variable that accomplishes this for us.

```
In [20]: LRLag = ps.lib.weights.lag_spatial(W, data1.Housholds_with_Electric_Lighting_Ratio)

In [21]: LRLagQ4 = ps.viz.mapclassify.Quantiles(LRLag, k=4)

In [22]: LRLagQ4
```

```
Out[22]:                Quantiles

         Lower          Upper            Count
         =====================================
                 x[i] <= 0.433               9
         0.433 < x[i] <= 0.521               9
         0.521 < x[i] <= 0.615               8
         0.615 < x[i] <= 0.722               9
```

```
In [23]: f, ax = plt.subplots(1, figsize=(20, 10))
         india.assign(cl=LRLagQ4.yb).plot(column='cl', categorical=True, \
                 k=4, cmap='OrRd', linewidth=0.1, ax=ax, \
                 edgecolor='white', legend=True)
         ax.set_axis_off()
         plt.title("Housholds with Electriciy Ratio Spatial Lag Deciles")

         plt.show()
```



The decile map for the spatial lag tends to enhance the impression of value similarity in space.

1.3.1 Local Autocorrelation Statistics

PySAL has many local autocorrelation statistics. Let's compute a local Moran statistic for the same data shown above

```
In [24]: LMo_HR90 = ps.explore.esda.Moran_Local(data1.Housholds_with_Electric_Lighting_Ratio.va
```

```
C:\ProgramData\Anaconda3\lib\site-packages\pysal\explore\esda\moran.py:895: RuntimeWarning: inv
  self.z_sim = (self.Is - self.EI_sim) / self.seI_sim
C:\ProgramData\Anaconda3\lib\site-packages\scipy\stats\_distn_infrastructure.py:879: RuntimeWar
  return (self.a < x) & (x < self.b)
```

```
C:\ProgramData\Anaconda3\lib\site-packages\scipy\stats\_distn_infrastructure.py:879: RuntimeWa
  return (self.a < x) & (x < self.b)
C:\ProgramData\Anaconda3\lib\site-packages\scipy\stats\_distn_infrastructure.py:1738: RuntimeW
  cond2 = (x >= self.b) & cond0
```

```
In [25]: sig = LMo_HR90.p_sim < 0.05
```

```
In [26]: sig.sum()
```

```
Out[26]: 9
```

```
In [27]: hotspots = LMo_HR90.q==1 * sig
         hotspots
```

```
Out[27]: array([False, False, False, False, False, False, False, False, False,
                False, False, False, False, False, False,  True,  True, False,
                False, False, False, False, False, False, False, False,  True,
                False, False, False,  True, False, False, False, False])
```

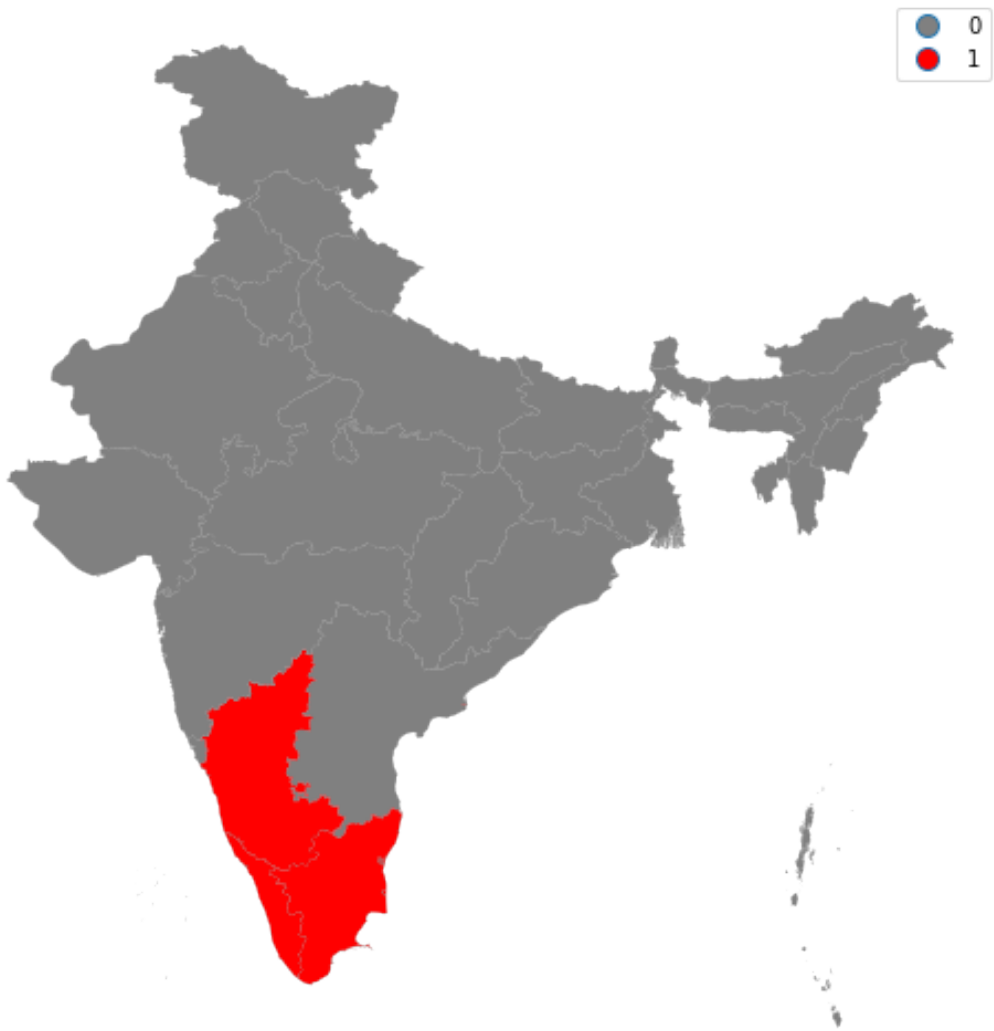
```
In [28]: coldspots = LMo_HR90.q==3 * sig
         coldspots
```

```
Out[28]: array([False, False, False, False,  True, False, False, False, False,
                False, False, False, False, False,  True, False, False,  True,
                False, False, False, False, False, False, False, False, False,
                False, False, False, False, False, False, False,  True])
```

1.3.2 HotSpots

HotSpots are the region which are similar and have higher ratio of household with electricty to total households.

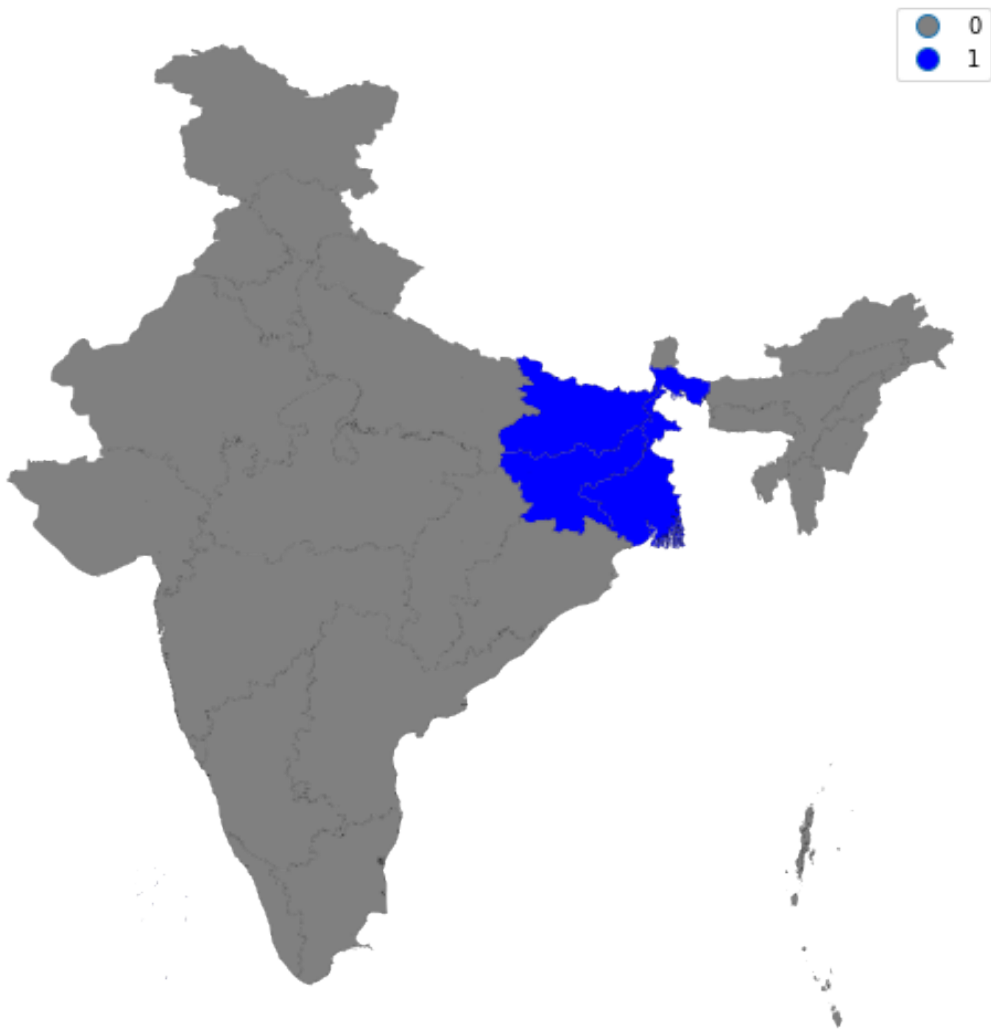
```
In [29]: from matplotlib import colors
         hmap = colors.ListedColormap(['grey', 'red'])
         f, ax = plt.subplots(1, figsize=(9, 9))
         india.assign(cl=hotspots*1).plot(column='cl', categorical=True, \
                 k=2, cmap=hmap, linewidth=0.1, ax=ax, \
                 edgecolor='grey', legend=True)
         ax.set_axis_off()
         plt.show()
```



1.3.3 ColdSpots

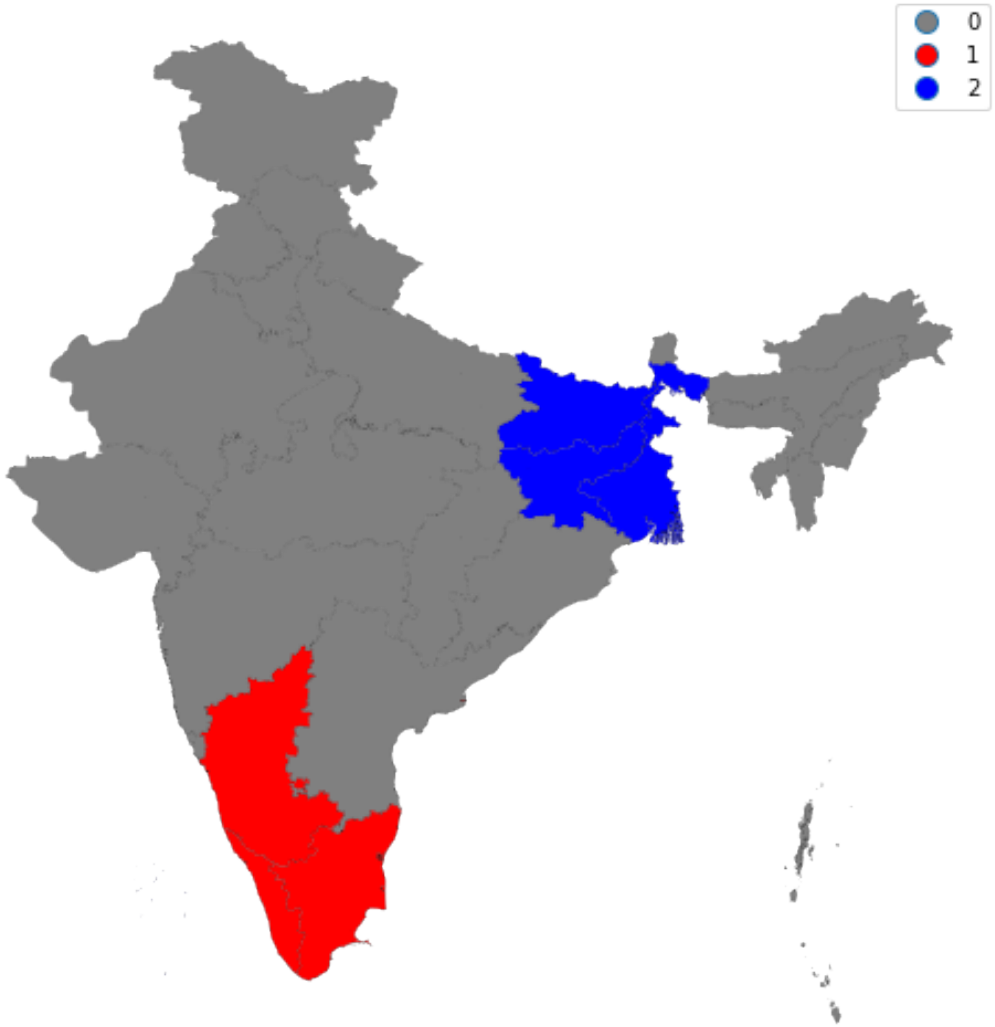
ColdSpots are the region which are similar and have lower ratio of household with electricty to total households.

```
In [30]: cmap = colors.ListedColormap(['grey', 'blue'])
         f, ax = plt.subplots(1, figsize=(9, 9))
         tx.assign(cl=coldspots*1).plot(column='cl', categorical=True, \
                 k=2, cmap=cmap, linewidth=0.1, ax=ax, \
                 edgecolor='black', legend=True)
         ax.set_axis_off()
         plt.show()
```



1.3.4 HotSpots and ColdSpots

```
In [31]: from matplotlib import colors
         hcmap = colors.ListedColormap(['grey', 'red','blue'])
         hotcold = hotspots*1 + coldspots*2
         f, ax = plt.subplots(1, figsize=(9, 9))
         tx.assign(cl=hotcold).plot(column='cl', categorical=True, \
                 k=2, cmap=hcmap,linewidth=0.1, ax=ax, \
                 edgecolor='black', legend=True)
         ax.set_axis_off()
         plt.show()
```



1.4 Reference

1. http://darribas.org/gds_scipy16/